Last Time:

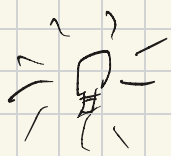
- Rigid-Body Dynamics
- Drones
- Airplanes
- Rockets
- Flying Cars

Today:

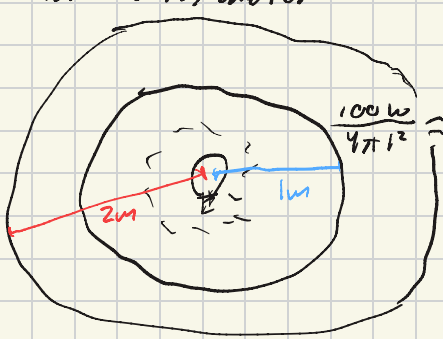
- Gravity + Inverse-Square Law
- Restricted Two-Body Problem
- Elliptical Orbits

Inverse-Square Law:

- Imagine a spherical 100W lightbulb!



- The light spreads out as it moves away from the light bulb. The ammount of illumination at a point is given by the flux in watts/meter^2



- At 1meter away from the bulb the power is spread out on a sphere of area $4\pi 1^2 m^2$. At 2meters $\rightarrow 4\pi 2^2 m^2$

$\Rightarrow$ The illumination we see decreases like $\frac{1}{r^2}$

* This same geometric picture applies to any field as it spreads out:

- Light
- Gravity
- Sound
- Radio

Newton's Law of Gravitation:

$$F = G\frac{m_1 m_2}{r^2}$$

- F: magnitude of gravitational force
- m_1, m_2: masses
- r^2: distance
- G: "Gravitational Constant" a.k.a. "Big G"

$\approx 6.674 \times 10^{-11}\ m^3/kg/s^2$

Restricted Two-Body Problem:

* Assumptions:

1) There are only 2 massive bodies
2) Both bodies are perfectly spherical
3) The second body is much less massive than the first body (e.g. a satellite orbiting ce planet)



* Newton's 2nd Law:

$$m_2 \ddot{\vec{r}} = \frac{-G m_1 m_2}{\vec{r} \cdot \vec{r}} \hat{r}$$

- Since m_2 (satellite mass) appears on both sides, it cancels out:

$$\ddot{\vec{r}} = \frac{-G m_1}{r^2} \hat{r}$$

- Since $G m_1$ always shows up together, we give it a name: "standard gravitational parameter":

$$\mu = G m_1 \Rightarrow \boxed{\ddot{\vec{r}} = \frac{-\mu}{r^2} \hat{r}}$$

- It turns out it's easier to measure μ than G or m_1 separately, so we know μ to much higher precision.

- This is the most basic model for the dynamics of a satellite orbit.

What Shape Are Satellite Orbits?

- The orbit will be confined to the plane spanned by the initial position ($\vec{r}_0$) and velocity ($\vec{v}_0$) since there are no forces acting out of the plane.

  $\Rightarrow$ Let's work in this 2D orbit plane

- We're going to re-write the dynamics in polar coordinates

- Let's write the Lagrangian:

$$T = \frac{1}{2} m v^T v = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$$

$$F = \frac{-\mu m}{r^2} \hat{r} \Rightarrow U = \frac{-\mu m}{r}$$ } doesn't depend on θ

$$\Rightarrow L = T - U = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) + \frac{\mu m}{r}$$

- Euler-Lagrange Equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = m\ddot{r} - mr\dot{\theta}^2 + \frac{\mu m}{r^2} = 0$$

$$\Rightarrow \boxed{\ddot{r} - r\dot{\theta}^2 + \frac{\mu}{r^2} = 0}$$ ← "radial equation"

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = 0 \Rightarrow \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} = \text{const.}$$ } conservation of angular momentum

($\frac{\partial L}{\partial \dot{\theta}}$: angular momentum)

$$\Rightarrow \boxed{h = r^2\dot{\theta} = \text{const.}}$$ ← "specific angular momentum"

- We can also write h in Cartesian coordinates:

$$h = r^2\dot{\theta} = x\dot{y} - y\dot{x} = \begin{bmatrix} -y \\ x \end{bmatrix}^T \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \left(\underbrace{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}_{S} \begin{bmatrix} x \\ y \end{bmatrix} \right)^T \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix}$$

- Notes on S:

* $S^T = -S$ (skew-symmetric)
* $S^2 = SS = -I$ (acts like imaginary i)
* $S^T S = I$
  ↳ 90° rotation matrix

* In 3D the full angular-momentum vector is conserved:

$$\vec{h} = \vec{r} \times \vec{v} = \text{const.}$$

- Our 2D scalar result is essentially the $\hat{z}$-component of this cross product if we choose coordinates where $\vec{r}$ and $\vec{v}$ are in the x-y plane.

- Look at the radial equation:

$$\ddot{r} - r\dot{\theta}^2 = \frac{-\mu}{r^2}$$

* This is a nonlinear 2nd-order ODE. Usually these can't be solved for $r(t)$.

- We will use some tricks to find $r(\theta)$ (shape/curve of orbit) instead $r(t)$.

* First Trick: use $h = \text{const.}$ to eliminate $\dot{\theta}$

$$\Rightarrow \ddot{r} - \frac{h^2}{r^3} = \frac{-\mu}{r^2}$$

* Second Trick: change variables to $u = \frac{1}{r}$:

$$\dot{r} = \frac{d}{dt} r = \frac{d}{dt}\left(\frac{1}{u}\right) = \frac{-1}{u^2}\frac{du}{dt}$$

* Third Trick: Change independent variable from t to θ:

$$\frac{d}{dt} = \frac{d}{d\theta}\dot{\theta} = \frac{h}{r^2}\frac{d}{d\theta} = hu^2\frac{d}{d\theta}$$

$$\Rightarrow \dot{r} = \frac{-1}{u^2}\frac{du}{dt} = -h\frac{du}{d\theta} \Rightarrow \ddot{r} = -h^2u^2\frac{d^2u}{d\theta^2}$$

* Plug this back into the original ODE:

$$-h^2u^2\frac{d^2u}{d\theta^2} - u^3h^2 = -u^2\mu \Rightarrow \boxed{\frac{d^2u}{d\theta^2} + u = \frac{\mu}{h^2}}$$

This is a linear 2nd-order ODE!

* Homogeneous Solution: $u_h = C_1\sin(\theta) + C_2\cos(\theta) = C\cos(\theta+\theta_0)$
* Particular solution: $u_p = \frac{\mu}{h^2}$

$$\Rightarrow \quad u(\theta) = \frac{\mu}{h^2} + C\cos(\theta) \qquad (\text{assume } \theta_0 = 0)$$

- Convert back to original coordinates:

$$r(\theta) = \frac{1}{u(\theta)} = \frac{1}{\frac{\mu}{h^2} + C\cos(\theta)} = \frac{h^2/\mu}{1 + e\cos(\theta)}$$

(e: "eccentricity")

This is the equation for a conic section!

* Cases:

$e = 0 \Rightarrow$ circle, $\quad 0 < e < 1 \Rightarrow$ ellipse

$e = 1 \Rightarrow$ parabola, $\quad e > 1 \Rightarrow$ hyperbola